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The association between age of menopause and type 2 diabetes: a systematic review and meta-analysis

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Abstract

Background Type-2 diabetes mellitus (T2DM) is considered one of the chronic diseases that can have a relationship with age of menopause onset. Several studies have revealed that early menopause or late menopause can have a correlation with type-2 diabetes. This systematic review and meta-analysis aimed to investigate the association between the age of menopause onset and type-2 diabetes.

Methods and materials PubMed, Web of Science, Scopus, Science Direct, Google Scholar, and Cochrane were searched for studies that have evaluated the relationship between T2DM and age of menopause onset. We pooled the effect sizes of the included studies using both adjusted odds ratios (OR) and hazard ratios (HR) of interest outcomes with their 95% confidence interval (CI).

Results Nineteen papers were included in this study, 8 studies were cohorts, and 11 were cross sectional. Studies revealed a statistically significant association between early menopause age and increased odds of T2DM (OR = 1.24, 95% CI: 1.09–1.40; I² = 67%; *p* = 0.001). Late menopause age was also associated with an increased odds of T2DM (OR = 1.14, 95% CI: 1.03–1.26; I² = 56%; *p* = 0.01) compared to the reference group with normal menopausal age. As our secondary outcome, the hazard of developing T2DM in individuals with early or late menopausal age was assessed. Pooled analysis demonstrated a significantly higher hazard of T2DM among women with early menopause age (HR = 1.31, 95% CI: 1.05–1.64; I² = 72%; *p* = 0.02). Late menopause age did not show a significant association (HR = 0.96, 95% CI: 0.84–1.10; I² = 68%; *p* = 0.56).

Conclusion Early and late menopause can both increase the risk of T2DM. Future research is needed to warrant the certainty of our findings.

Keywords Premature-menopausal, Post-menopausal, Type-2 diabetes mellitus

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Introduction

Menopause is defined as the lack of a menstrual period for at least a year [1]. Menopause is known as a multidimensional physiological event that has significant effects on the risk of other diseases and quality of life [2]. The age of menopause onset is one of the most important determinants of women's health in the future [3, 4]. The basic mechanisms of the onset of menopause are not yet fully understood, and the age of onset of menopause depends on complex factors [5]. Therefore, the age of onset of menopause varies in different parts of the world. For example, in Western women, normal menopause typically occurs between the ages of 40 and 60, with an average age of onset of 51 [6], but the average age of onset of menopause in Asian, African, Australian, and Middle East is different [7]. According to the World Health Organization (WHO), by 2030, there will be an estimated 1.2 billion post-menopausal women worldwide, with 47 million women going through menopause each year [8]. In the United States, more than 11% of women over 20 have diabetes. It is estimated that one in three girls born in 2000 will develop diabetes during their lifetime [9].

Age of menopausal onset is an important determining factor in women's health outcomes [10]. The age of onset of menopause may significantly impact morbidity and mortality [11]. Numerous studies have shown that several factors affect the age of menopause; these factors include fertility factors, demographics, socioeconomic, lifestyle, and cultural factors [12]. Menopausal age may be associated with many diseases, including diabetes, gastroenteritis, chronic kidney disease, CVD and hypertension, and hyperlipidemia [13, 14]. Some chronic diseases, including diabetes, lead to reproductive aging and may lead to premature aging of the ovaries [15, 16].

There have been conflicting articles on the relationship between menopausal age and type-2 diabetes. Therefore, this systematic review and meta-analysis aimed to investigate the possible relationship between the age of menopause and type-2 diabetes by quantitatively aggregating the available evidence.

Methods

This study was carried out according to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) statement [17]. The protocol of this study in PROSPERO, with code CRD42017080789.

Search strategy

Articles published in the electronic databases of PubMed Web of Science, Scopus, Science Direct, Google Scholar, and Cochrane were performed using MESH-based keywords from 2000 to 21 December 2022 without language

restrictions. Reference lists, key journals, and gray literature were also used in the search.

Inclusion and exclusion criteria

The specific criteria for inclusion were as follows: (i) Studies carried out on women who had reached menopause, and (ii) Studies that provided data that could be extracted regarding the possible relationship between early or late menopause age and risk of type 2 diabetes mellitus (T2DM). Both cohort studies and case-control studies were considered. Studies were excluded if they: (i) lacked a control group consisting of individuals without T2DM, (ii) Encompassed pre- or peri-menopausal women, (iii) Involved participants receiving concurrent therapy with medications that affect glucose metabolism, such as glucocorticoids, (iv) Were case reports, case series, review articles, and (v) Were conducted on animals.

Quality assessment

The methodological quality and risk of bias of the included studies were independently assessed by two reviewers using appropriate tools, such as the Newcastle-Ottawa Scale for observational studies (NOS). NOS is designed to evaluate the methodological quality of articles to assess the risk of bias for all three categories of study, i.e., cohort, case, and cross-sectional. This tool consists of three sections: selection process, comparability, and outcome measurement. Studies with 7–9, 5–6, 1–4 were rated good, fair, and poor, respectively.

Study selection and data extraction

Two independent reviewers screened the titles and abstracts of the identified studies for potential inclusion. Full-text articles of potentially relevant studies were then assessed for eligibility. Any discrepancies between the reviewers were resolved through discussion or consultation with a third reviewer. Study characteristics, including the first author's name, year of publication, sample size, type of study, Menopause classification, Duration of study, Covariates and the quality of the studies were extracted in an Excel spreadsheet.

Outcomes

The main objective of this study is to examine the association between age at menopause and the odds of developing type 2 diabetes mellitus (T2DM), comparing early and late menopausal ages to the reference group with normal menopausal age. Additionally, we will assess the hazard of developing T2DM among individuals with early or late menopausal age as secondary outcomes.

Data synthesis

We pooled the effect sizes of the included studies using both adjusted odds ratios (OR) and hazard ratios (HR) of interest outcomes with their 95% confidence interval (CI). For adjusted effect sizes, we used the fully adjusted model presented in each article. For studies reported the OR of having T2DM, we used the lowest and the highest quartile in case of early menopause and late menopause, respectively. A funnel plot and Egger's regression test were used to assess possible sources of publication bias for outcomes with at least 10 studies included. Sensitivity analysis was performed using the leave-one-out method for our primary outcomes.

Results

After removing duplicates, the screening process was conducted on 2412 papers, assessing 62 articles for eligibility. Finally, 19 papers were included in this study [10, 18–35] (Fig. 1). It should be noted that 8 studies were cohorts, and 11 were cross-sectional. Characteristics of

included studies and quality assessment are summarized in Table 1.

Primary outcomes

The pooled analysis of the included studies revealed a statistically significant association between early menopause age and increased odds of T2DM (pooled odds ratio [OR]=1.24, 95% confidence interval [CI]: 1.09–1.40; $I^2=67\%$; $p=0.001$) (Fig. 2). Publication bias was assessed using funnel plots, which appeared relatively symmetrical. Furthermore, Egger's regression test did not show any statistical evidence of publication bias ($p=0.11$). Sensitivity analyses were performed by excluding one study at each time. The results indicated that the overall findings were robust and consistent (Supplementary file).

In addition, late menopause age was also associated with an increased odds of T2DM (pooled OR=1.14, 95% CI: 1.03–1.26; $I^2=56\%$; $p=0.01$) compared to the reference group with normal menopausal age (Fig. 3). Publication bias was assessed using funnel plots, which appeared relatively symmetrical (Fig. 4). Furthermore, Egger's

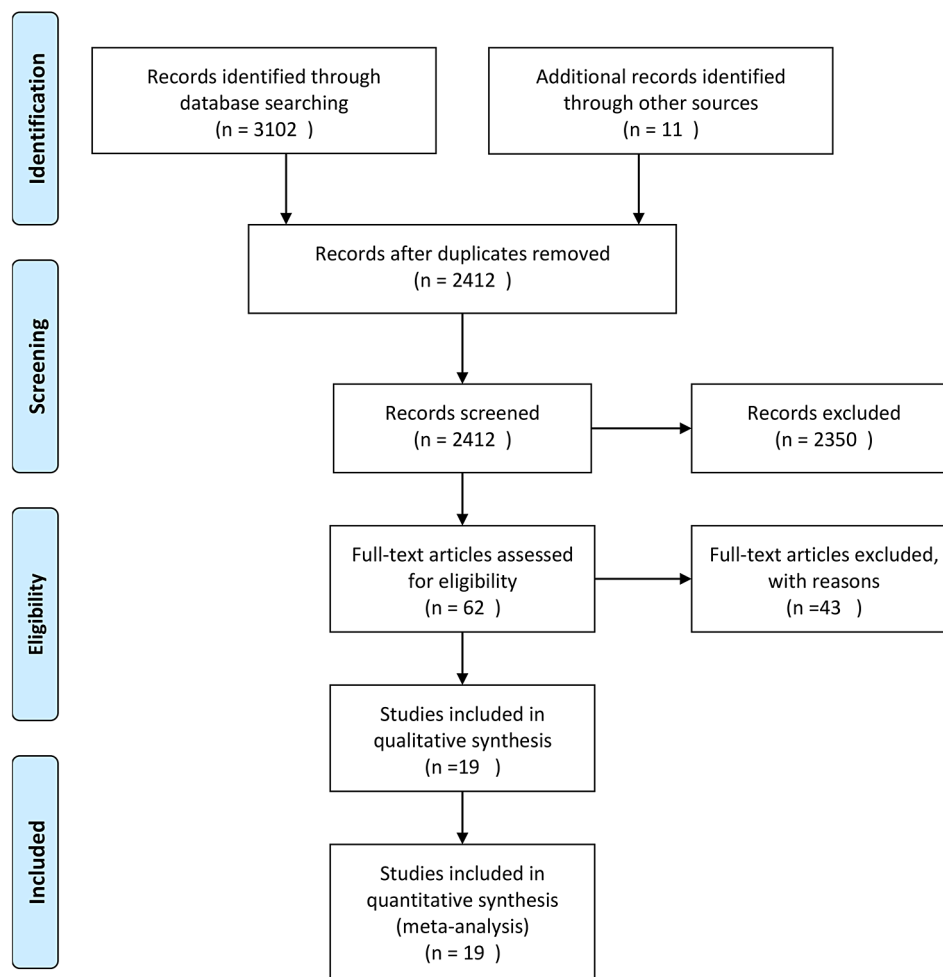


Fig. 1 PRISMA flow diagram

Table 1 Characteristics of the included studies

	Author	Year	Country	Type of Study	Total Participants	Menopause classification	Duration of study	Covariates	Quality (NOS)
1	Appiah et al.	2014	USA	Prospective cohort	2,597	NR	9.2 years	Age, race, education, smoking, physical activity, general vascular health status (HT), reproductive factors (parity, postmenopausal estrogen use, age at menarche, age at menopause, BMI, WC	Good
2	Aslanaj, Eralda	2019	Netherlands	Prospective Cohort	3,623	early (≤ 44 y old), normal (45–54 y old), and late (≥ 55 y old)	11 years	age, cohort, smoking, alcohol, education, physical activity, oral contraceptive use, hormone therapy, age at menarche, number of pregnancies	Good
3	Binh, Tran Quang	2016	Vietnam	Cross-sectional	608	early menopause < 45 years of age late menopause > 50	N.R	socio-economic status (income level, occupation, educational level, and marital status) and lifestyle-related factors (smoking, alcohol consumption, playing sport, and relax time)	Fair
4	Brand, J. S.	2013	10 western European countries, Netherlands	Prospective Cohort	7,864	< 40, 40–44, 45–49, and ≥ 55 years, menopausal age between 50 and 54 years taken as a reference	11 years	Age, BMI, smoking, alcohol, physical activity, HRT education, no. of pregnancies, oral contraceptive use	Good
5	Fu, Yingli	2016	China	Cross-sectional	2,099	early menopause = before 45, Late menopause = after 54, normal menopause = between 46–53	1 year	Anemia, Diabetes, Hyperlipidemia, Hypertension, Ischemic heart disease, COPD, CGPU, Cerebrovascular disease, Cholecystitis/gallstone Arthritis, Chronic low back pain, Poor mental health, Any disease (defined as one or more of the 11 chronic diseases or poor mental health)	Good
6	LeBlanc, Erin S	2017	USA	Prospective Cohort	124, 379	the shortest (< 30 years) reproductive periods, with medium length reproductive periods (36 to 40 years) and the longest (45 + years) reproductive periods	5 years	age, race, BMI, waist circumference, education, marital status, physical activity, smoking status, alcohol use, hormone therapy (HT) intervention arm, postmenopausal HT use, oral contraceptive use, metformin use, family history of diabetes, years since final menstrual period, and number of fullterm pregnancies	Good
7	Li, Y.	2020	China	Cross-sectional	5,693	early menopause (< 45 years), late menopause (> 54 y), normal menopause(45–54)	N.R	Age	Fair

Table 1 (continued)

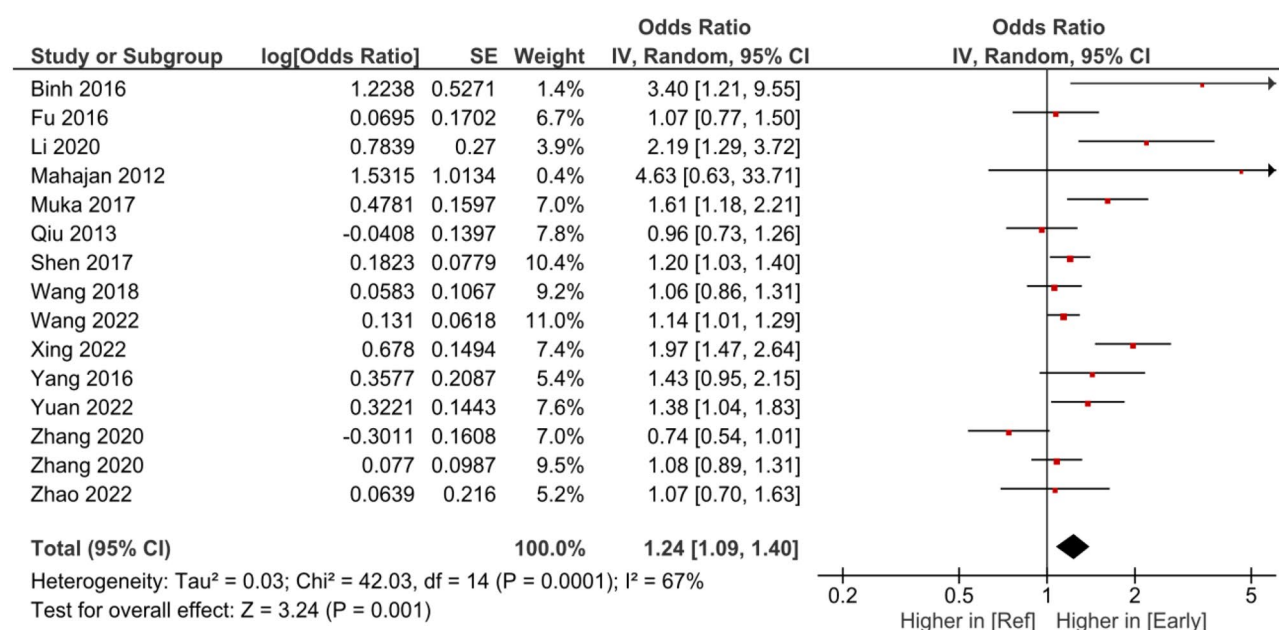
	Author	Year	Country	Type of Study	Total Participants	Menopause classification	Duration of study	Covariates	Quality (NOS)
8	Mahajan, N.	2012	India	Cross-sectional	100	< 40 years, 40–45 years, 45–50 years, ≥ 50 years	1 year	Not Adjusted	Fair
9	Muka, Taulant	2017	Netherlands	Prospective Cohort	5,639	premature menopause, < 40 years; early menopause, 40–44 years, normal menopause, 45–55 years; and late menopause, > 55 years	9.2 years	Age, HRT, age at menarche, number of pregnancies, BMI, glucose, insulin, TC, use of lipid-lowering medication, SBP, antihypertensive medications, alcohol, smoking, education, CVD, physical activity, CRP	Good
10	Pandeya, Nirmala	2018	Multiple countries	Prospective Cohort	126,721	occurring before age 40, 40–44, 45–49 and + 50 years (natural menopause defined as occurring before the age of 45 years)	9 years	Age, education, smoking, age at menarche, number of children, age at first birth, menopausal status or timing, HRT, BMI	Good
11	Qiu, Changsheng	2013	China	Cross-sectional	3,304	(≤ 46, 47–49, 50, 51–52, and 53+ years)	N.R	age at enrollment, physical activity, parity, smoking, alcohol consumption, family history of diabetes, age at menopause, type of menopause, BMI and waist circumference	Good
12	Shen, L	2017	China	Cross-sectional	16,299	early menopause (at age ≤ 45 years), 46–52, ≥ 53	N.R	age, educational level, marital status, smoking, drinking, physical activity, family history of diabetes, parity, and history of oral contraceptive and/or HRT use anthropometric and biochemical factors, including BMI, WHR, hypertension, HDL and LDL cholesterol, total cholesterol and TG levels	Good
13	Wang, M.	2022	China	Prospective cohort	41,789	less than 40 (premature menopause), 40–44 (early menopause), 45–49 (as reference), 50–53, and 54 year or older (later age at menopause).	10.8 years	education, household income, health behaviors of smoking, alcohol drinking, physical activity, and anthropometric measurements including body mass index, waist circumference at baseline, health status of hypertension, and family history of diabetes, reproductive factors of age at menarche, number of live births, age at first birth, breastfeeding duration per child, and OC use	Good

Table 1 (continued)

	Author	Year	Country	Type of Study	Total Participants	Menopause classification	Duration of study	Covariates	Quality (NOS)
14	Wang, Meng	2018	China	Cross-sectional	17,076	≤ 44, 45–52 and ≥ 53 years	4years	age, marriage status, education level, occupation, household income, parity, family history of diabetes, smoking status, alcohol consumption, sedentary/physical activity and use of contraceptives, physical measurements of BMI and waist circumference, hypertension	Good
15	Xing, Z.	2022	USA	Cross-sectional	4,968	early menopause (40–44 years), later menopause (over 55 years), normal menopause (45–54)	2011–2018	age, race, education, BMI, smoking, hypertension, high cholesterol, physical activity, and family history of diabetes	Good
16	Yang, Aimin	2016	China	Cross-sectional	5063	years since menopause (≤ 5, 6–10, 11–15, 16–20, ≥ 21 years)	June2011 to December 2013	age, education, marital status, BMI, occupation, tobacco smoke, alcohol drinking, times of delivery, age at first delivery abnormal lipid, hypertension, and family history of diabetes	Good
17	Yuan, Y.	2022	China	Cross-sectional	5090	< 44, 44–52, > 52 years	N.R	age, education level, marital status, annual family income, family history of diabetes mellitus, drinking frequency, smoking status, physical activity, use of oral contraceptive and number of live births; hypertension, dyslipidemia, BMI, abdominal obesity; and the mutual adjustment of age at menarche and menopause.	Good
18	Zhang, L.	2020	China	Prospective cohort	15,406	early menopause with menopausal age ≤ 43 years; normal menopause with menopausal at 44–53 years; later menopause with menopausal age ≥ 54 years.	N.R	age, education level, average monthly individual income, marital status, smoking, alcohol drinking, physical activity and family history of diabetes, age at menarche, parity, the cause of the menopause, use of oral contraceptive pills, coronary heart disease, stroke and dyslipidemia	Good

Table 1 (continued)

Author	Year	Country	Type of Study	Total Participants	Menopause classification	Duration of study	Covariates	Quality (NOS)
19 Zhao, Y.	2022	China	Cross-sectional	4,279	early menopause (< 43 y), late menopause (> 54 y), normal menopause (44–53 y)	June 1, 2011 to November 30, 2011	age, marital status, education level, occupation, smoking status, alcohol consumption, family history of T2DM, physical activity, sleep quality, use of oral contraceptive pills, parity, and age at menarche, BMI, WC, stroke, coronary heart disease, and dyslipidemia	Good

**Fig. 2** Results of the adjusted odds ratios (95% confidence intervals) meta-analysis for type 2 diabetes mellitus according to early menopause age

regression test did not show any statistical evidence of publication bias ($p=0.26$). Sensitivity analyses were performed by excluding one study at each time. The results indicated that the overall findings were robust and consistent (Supplementary file).

Secondary outcomes

As our secondary outcome, the hazard of developing T2DM in individuals with early or late menopausal age was assessed. Among the included studies that reported hazard ratios (HRs), the pooled analysis demonstrated a significantly higher hazard of T2DM among women with early menopause age (pooled HR=1.31, 95% CI: 1.05–1.64; $I^2=72\%$; $p=0.02$). Conversely, late menopause age did not show a significant association (pooled HR=0.96, 95% CI: 0.84–1.10; $I^2=68\%$; $p=0.56$) (Fig. 5).

Quality assessment

The methodological quality and risk of bias of the included studies were assessed using appropriate tools, such as the Newcastle-Ottawa Scale for cohort studies and case-control studies. The majority of the included studies were of high quality, with low risk of bias in most domains assessed (Supplementary file).

Discussion

The present systematic review and meta-analysis aimed to investigate the association between age at menopause onset and the risk of developing type 2 diabetes mellitus (T2DM). The results from the pooled analysis of the included studies provided valuable insights into this relationship.

Our findings demonstrated a significant association between early menopause age and an increased odds of T2DM. Women who experienced menopause at an

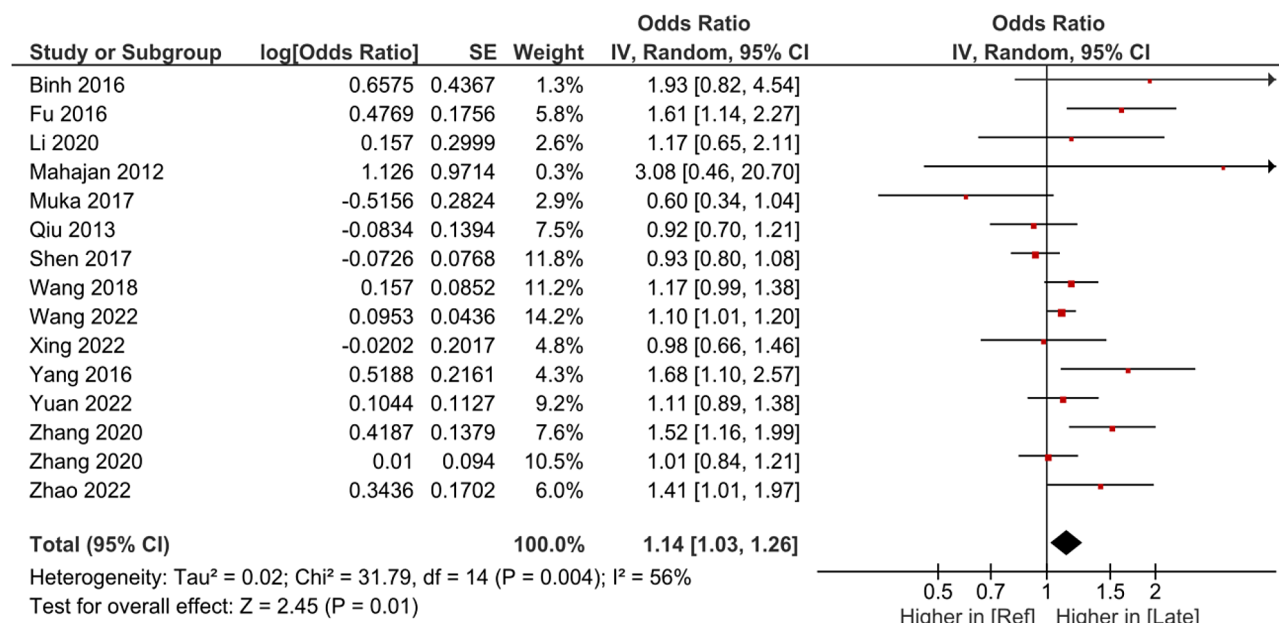


Fig. 3 Results of the adjusted odds ratios (95% confidence intervals) meta-analysis for type 2 diabetes mellitus according to late menopause age

earlier age had 24% higher odds of developing T2DM compared to those with a normal menopausal age. This observation is consistent with previous studies that have reported a link between early menopause and increased T2DM risk [36]. The underlying mechanisms for this association may be attributed to hormonal changes during menopause, including a decline in estrogen levels, which can affect insulin sensitivity and glucose metabolism [37].

Furthermore, early menopause is often associated with other risk factors for T2DM, such as increased body mass index (BMI) and unfavorable lifestyle behaviors [38]. These findings highlight the importance of considering early menopause as a potential risk factor for T2DM and the need for targeted preventive measures in this population. Interestingly, our results also indicated that late menopause age was associated with an increased odds of T2DM compared to the reference group with normal menopausal age. Women with a later onset of menopause had 14% higher odds of developing T2DM. This finding is somewhat unexpected, as late menopause has often been considered a protective factor against chronic diseases. However, it is important to interpret this result cautiously, as the magnitude of the association was relatively small and the heterogeneity among the included studies was moderate. Further research is needed to explore the underlying mechanisms and confirm this observation.

Regarding the secondary outcomes, our analysis assessed the hazard of developing T2DM in individuals with early or late menopausal age. The results revealed a significantly higher hazard of T2DM among women with early menopause age, indicating that they are at a greater

risk of developing T2DM over time. This finding suggests that early menopause may have long-term implications for metabolic health. On the other hand, late menopause age did not show a significant association with the hazard of T2DM. However, it is important to note that the evidence for this outcome was less robust, because of the low number of studies included in the analysis. Further investigations are warranted to provide more definitive conclusions on the relationship between late menopause and T2DM hazard.

There are several systematic reviews on this topic [36, 39, 40]. The first study by Anagnostis et al. [40] explores the relationship between early menopause, premature ovarian insufficiency (POI), and the risk of developing type 2 diabetes. The findings of the meta-analysis, in line with our results, revealed a significant association between early menopause and an increased risk of type 2 diabetes. Women who experienced early menopause were found to have 15% higher odds of developing type 2 diabetes compared to those who underwent menopause at the usual age. Similarly, women with premature ovarian insufficiency had 50% increased risk of type 2 diabetes compared to those without this condition.

In a dose-response meta-analysis conducted by Guo et al. [36], it was found that for every 5-year increase in age at menopause, there was a significant 10% reduction in the risk of developing type 2 diabetes in the future. These findings differ from our own study, which indicated an increased odds of type 2 diabetes among individuals with a later age of menopause. However, it is important to note that the results reported by Guo et al. may be limited due

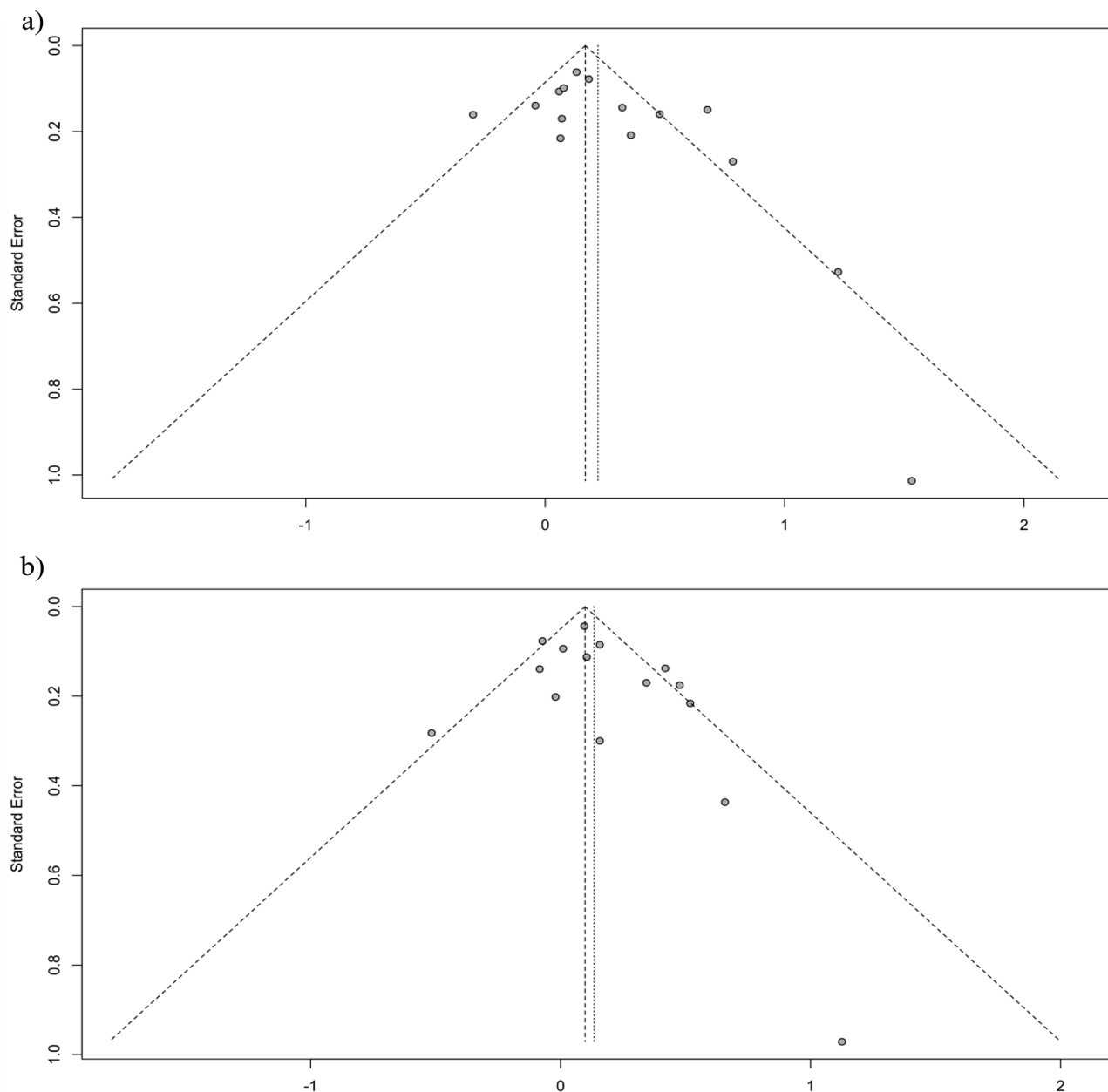


Fig. 4 Results of the funnel plot **a)** odds at early, **b)** odds at late menopause

to the inclusion of a relatively small number of studies, with only six studies being included in their analysis.

The most recent study by Liu et al. comprehensively evaluated the cardiometabolic diseases among individuals with early menopause [39]. The study analyzed data from 921,517 participants in 20 cohort studies conducted between 1998 and 2022. The findings showed that women who experienced premature menopause (PM) or early menopause (EM) had a higher risk of certain health conditions compared to women who experienced menopause at a later age (>45 years). Women with PM or EM had a higher risk of type 2 diabetes, hyperlipidemia,

coronary heart disease, stroke, and total cardiovascular events. However, there was no significant difference in the risk of hypertension between PM or EM women. Additionally, they found that PM women were associated with an increased risk of both ischemic and hemorrhagic stroke, while this association was not observed in EM women. However, the conclusion that both PM and EM women had a higher risk of total stroke was not supported by the findings.

While our study provides valuable insights into the association between age at menopause and T2DM, several limitations should be considered. First, the included

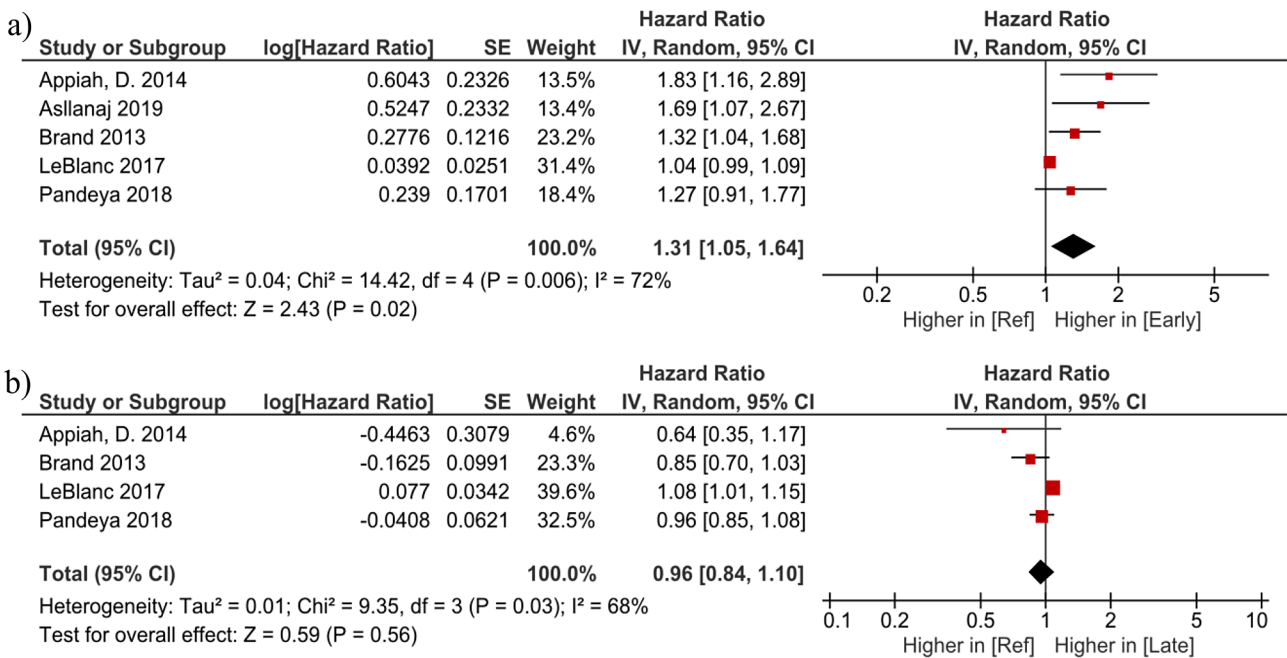


Fig. 5 Results of the adjusted hazard ratios (95% confidence intervals) meta-analysis for type 2 diabetes mellitus according to age at menopause **a)** early, **b)** late

studies were observational in nature. Second, there was considerable heterogeneity among the studies, potentially influenced by variations in study design, participant characteristics, and methodological approaches. Future studies on this topic are warranted in order to strength the evidence regarding late menopausal age and T2DM.

Conclusion

This systematic review and meta-analysis demonstrated that both early and late menopause ages are associated with an increased odds of developing T2DM. Women who experience menopause at an earlier age appear to be at a higher risk of T2DM, while late menopause age may also confer a modest increase in risk. These findings emphasize the importance of considering menopausal age as a potential risk factor for T2DM and highlight the need for further research to elucidate the underlying mechanisms and develop targeted preventive strategies. Clinicians should be aware of these associations and consider incorporating menopausal age as a factor in the assessment and management of T2DM risk in women.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12986-024-00858-0>.

Supplementary Material 1

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Not applicable.

Author contributions
K.K & M.B & M.Y: Conceptualization, Project Administration, Data curation, Writing- Original Draft, Writing – Review & Editing, Visualization A.S & K.J: Validation, Resources, Methodology, Software, Formal analysis, Writing – Original Draft S.PO&T.B &Z.M: Writing- Original Draft, Writing – Review & Editing S.E& M.P: Data curation, Writing – Review & Editing.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
This study was approved by ethical code IR.ABZUMS.REC.1399.203.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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